CSE 360: Networking Experiments Overview

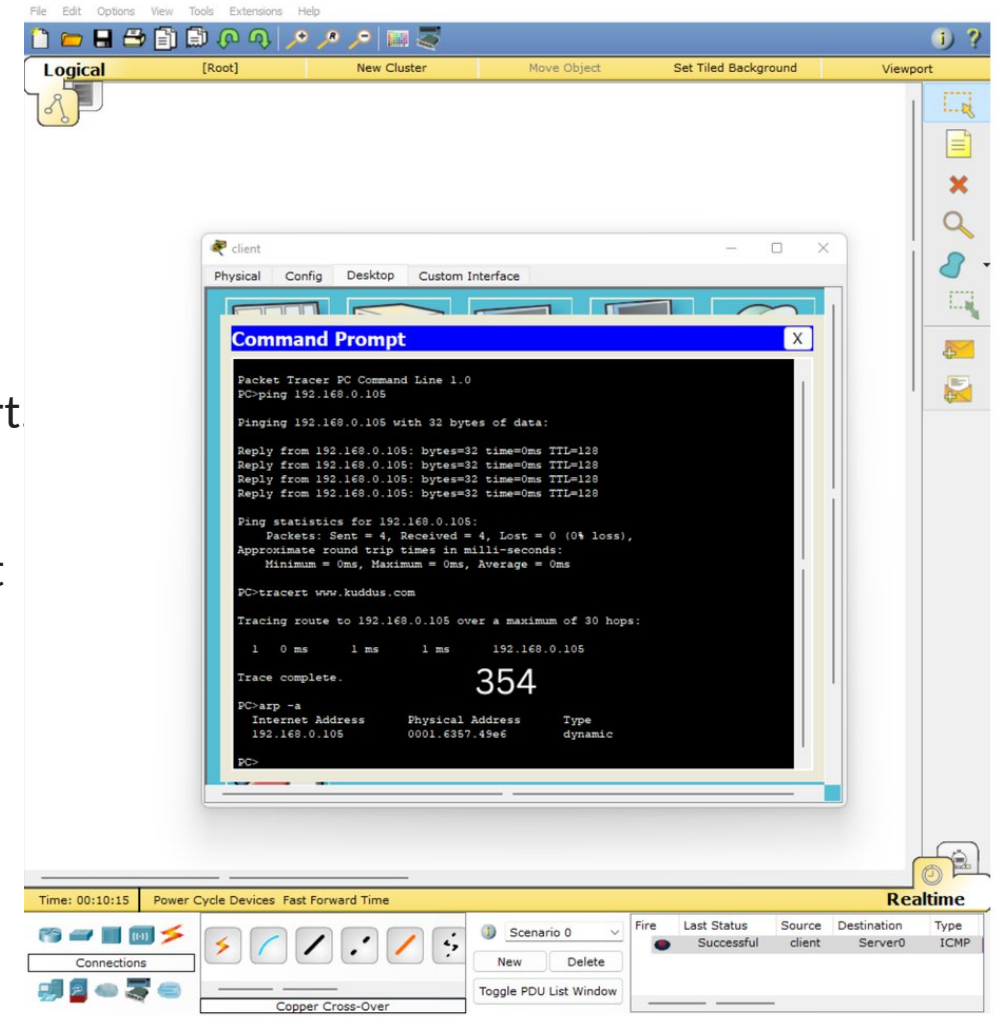
- Short walkthrough of the labs in slides 2, 4, 5, 6 and 7.
- Each section pairs the key ideas with results from my own Packet Tracer work.
- Concepts: LANs, device access, DNS and web, static routing, serial links, VLANs.

How the labs fit together

- Slides 3-5: local networks, device access, DNS and web (from slide set 2).
- Slides 6-7: static routing with two routers, plus the hub variant (from slide set 4).
- Slide 8: three routers in series (from slide set 5).
- Slide 9: mixed Ethernet and serial network (from slide set 6).
- Slide 10: VLANs with router-on-a-stick, static and DHCP addressing (from slide set 7).
- Slide 11: the check toolkit used in every lab.

Same-subnet LANs: hub vs switch vs wireless

- Same mask 255.255.255.0 means same network, e.g. all 192.168.0.x talk with no router.
- Hub repeats everything to all ports; one conversation at a time, collisions possible.
- Switch learns MAC addresses and forwards only to the right port
- Wireless is the same subnet idea without cables.
- Wait for green link lights (switch ports need ~30-60 s), then test with ping.



Talking to devices: console, passwords, Telnet, SSH

- Console cable + Terminal gives Switch> prompt directly on the box.
- Modes: user (>) , privileged (#) with enable, config with conf t.
- enable secret = password for > to #; line console 0 + login = console password.
- line vty 0 4 = 5 remote Telnet slots at once; SSH (PuTTY) is the encrypted equivalent.
- copy run startup-config (wr) saves, otherwise a reboot loses the work.

```
        --- System Configuration Dialog ---

Continue with configuration dialog? [yes/no]: no


Press RETURN to get started!


Router>en
Router#CONF T
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#conf t
%Invalid hex value
Router(config)#int fa 0/0
Router(config-if)#ip add 192.168.1.2 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int fa 0/1
Router(config-if)#ip add 192.168.3.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

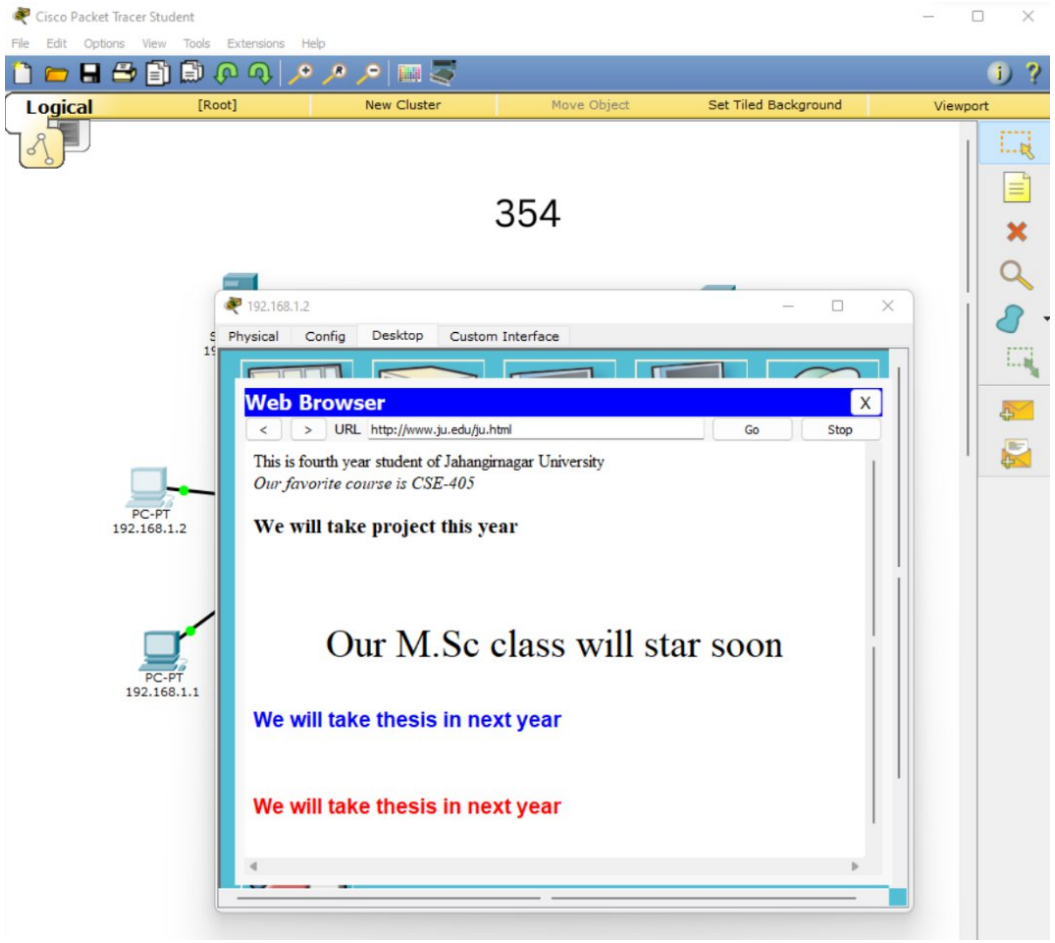
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
```

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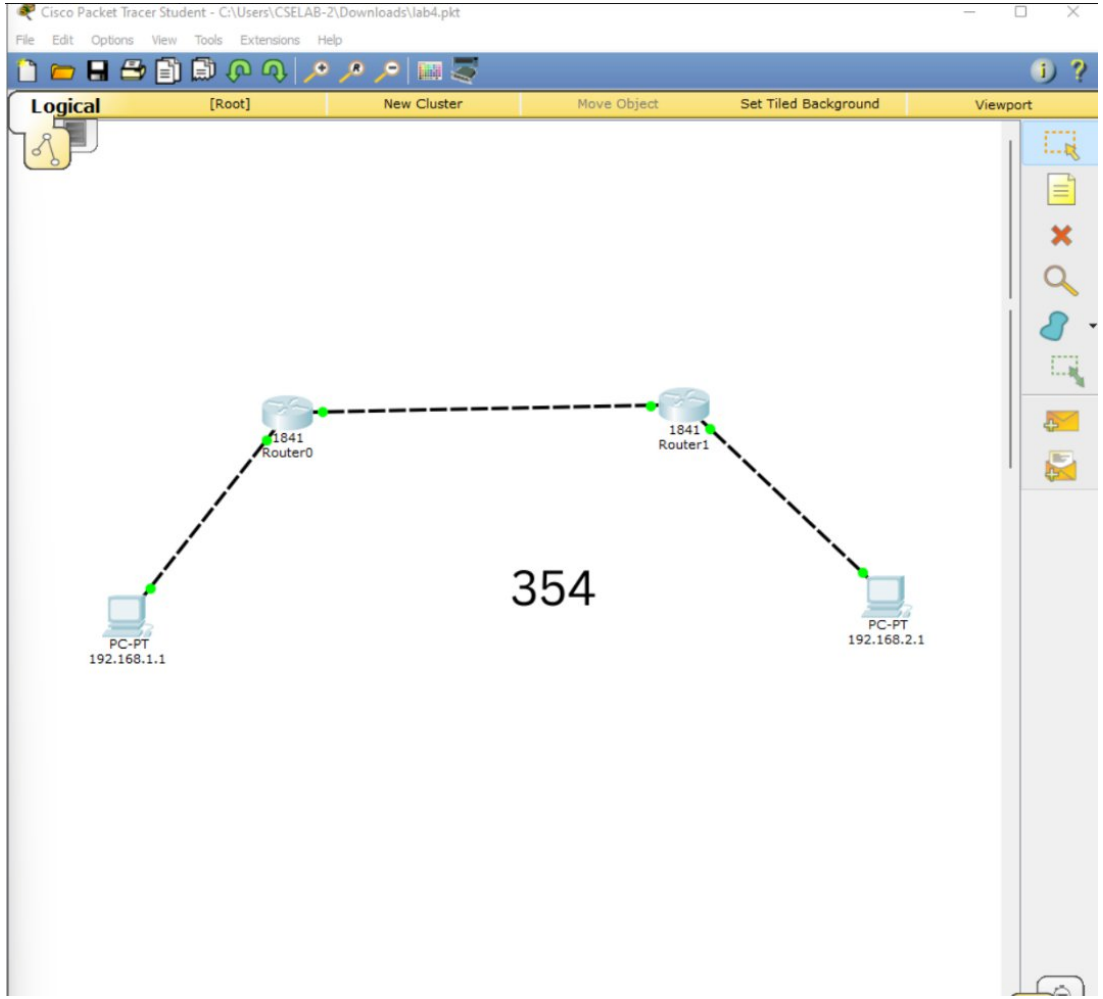
Client, DNS and web: JU / DU networks

- PC fields: IP + mask + gateway (local router leg) + DNS server (which server to ask).
- DNS A Record maps a name to an IP, e.g. www.ju.edu to 192.168.1.3. DNS Service must be On.
- HTTP service hosts ju.html / du.html. URL = name + file, e.g. http://www.ju.edu/ju.html.
- Router Fa0/0 192.168.1.4 and Fa0/1 192.168.2.4, both Port Status On.



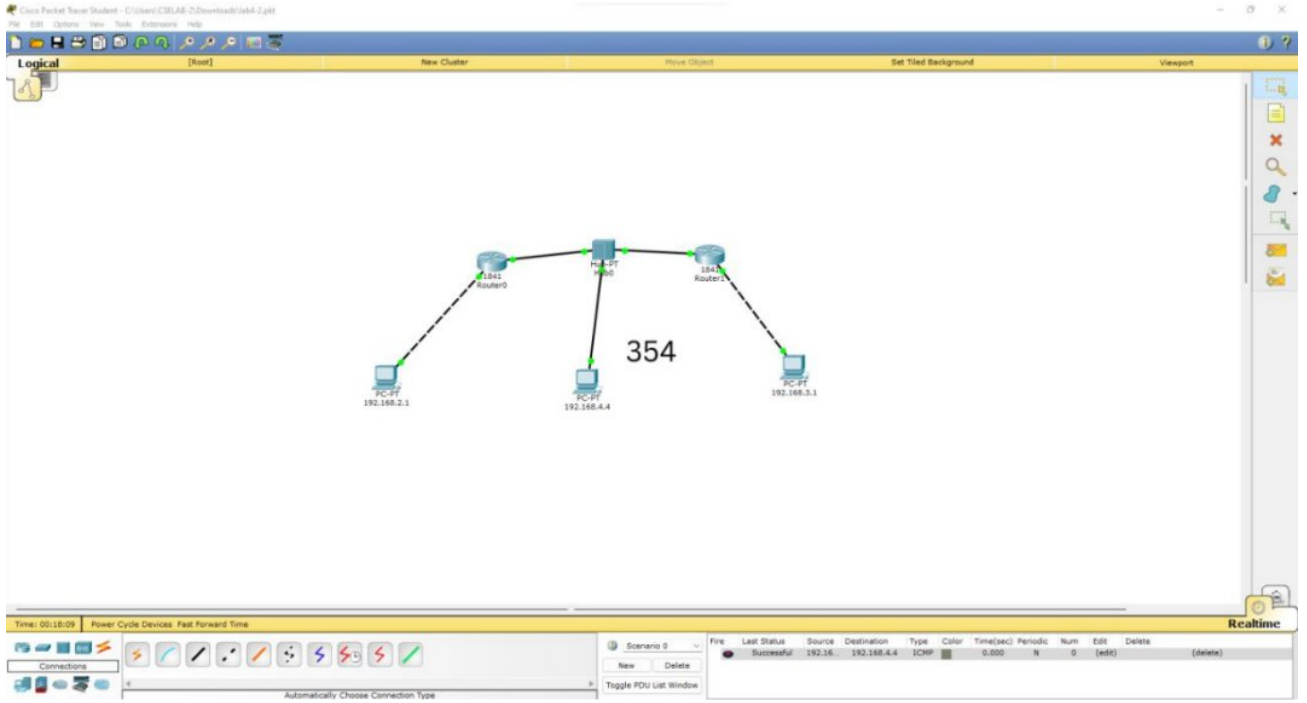
Static routing with two routers

- Three /24 networks: left LAN, middle router-to-router net, right LAN.
- Each interface: ip address + no shutdown (GUI: Port Status On).
- Connected routes (C) appear alone; remote nets need ip route <net> <mask> <next-hop>.
- Next hop must be directly reachable, e.g. 192.168.2.0 via 192.168.3.2.
- Read sh ip route: C = connected, S [1/0] = static. ping !!!!! = 5 replies.



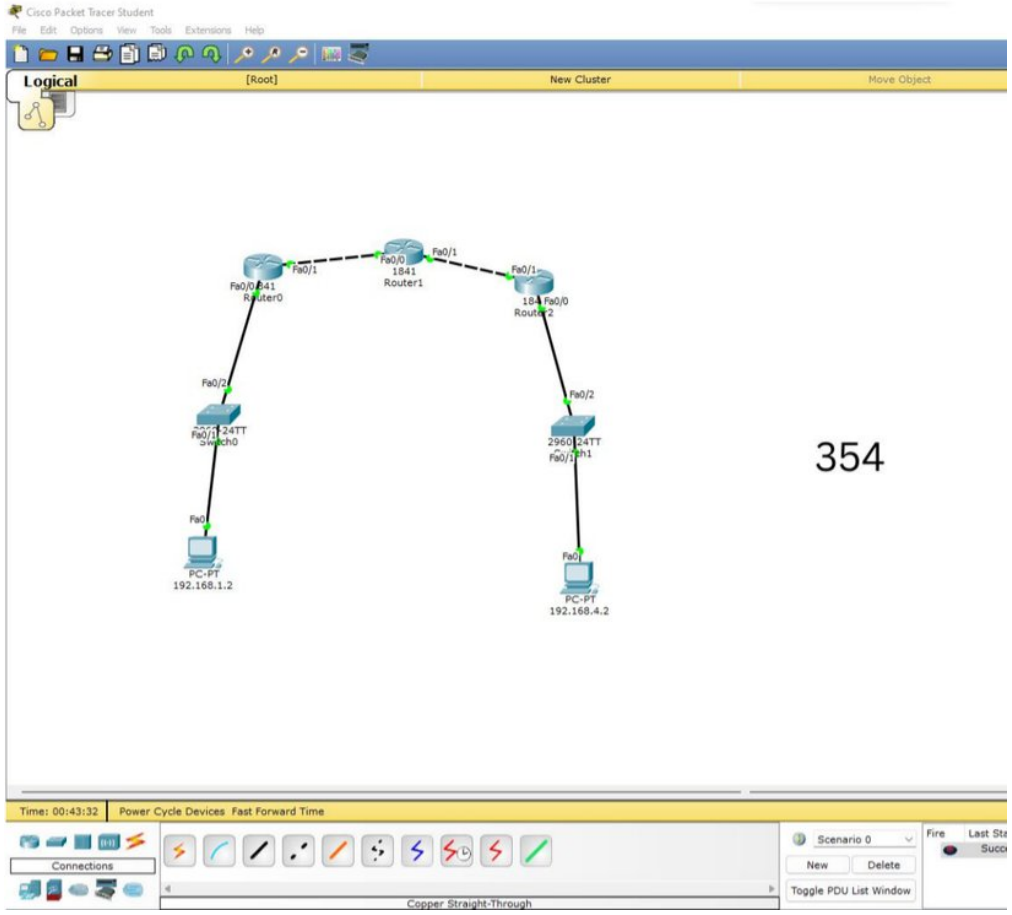
Variant: a host on a hub between routers

- Same routing idea; the new part is the shared middle subnet.
- Hub = one shared medium: Router0 leg + middle PC + Router1 leg all in 192.168.4.0/24.
- Each router still needs one static route to the far end only.
- Middle PC reaches both sides with no extra routes (it is directly connected).
- Same port rule: every used interface Port Status On.



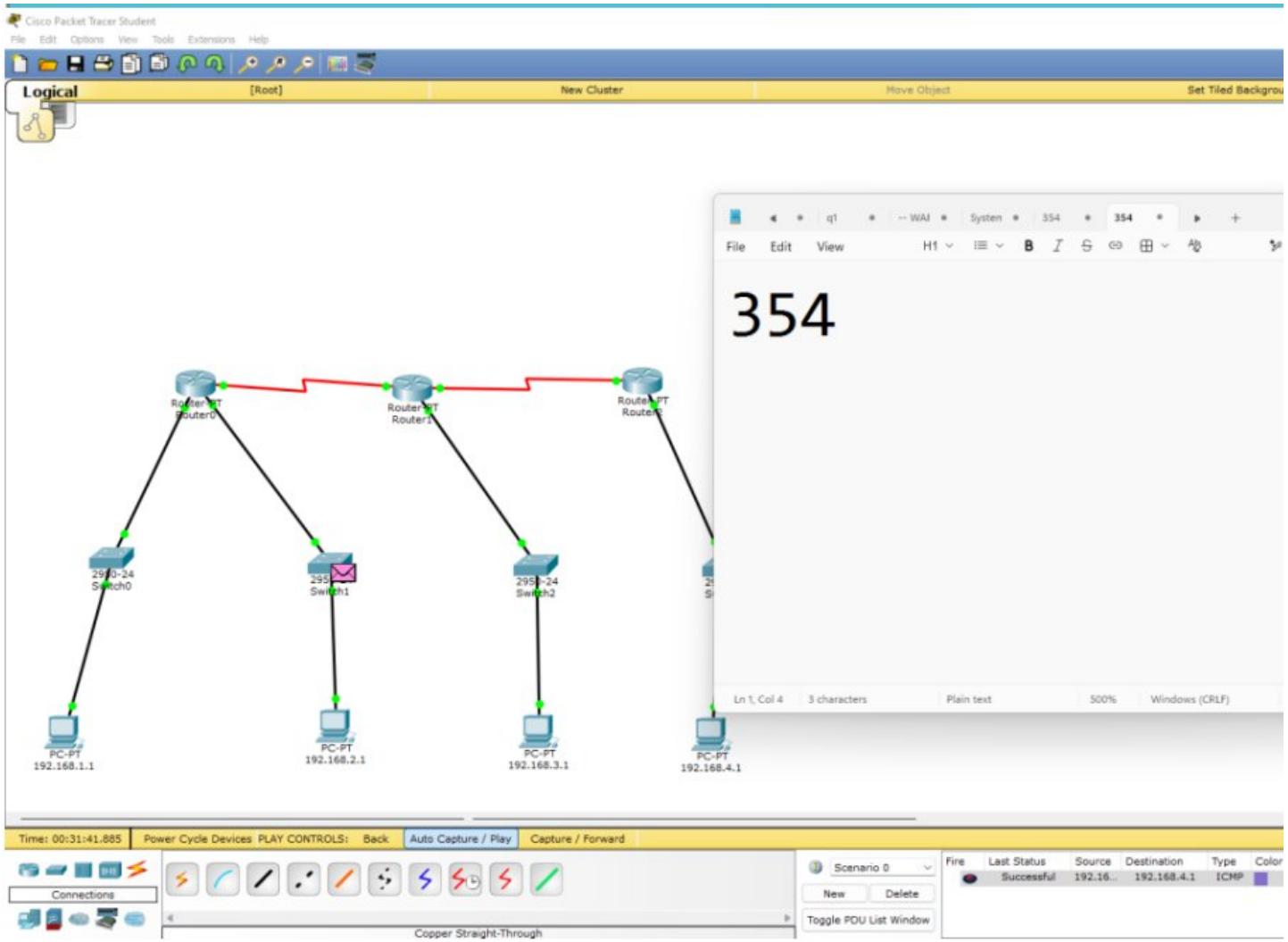
Three routers in series

- Four /24 networks end to end: 192.168.1.0 - .2.0 - .3.0 - .4.0.
- PC0 192.168.1.2 gw .1.1; PC1 192.168.4.2 gw .4.1.
- Each router learns 2 networks alone, so 2 static routes each (6 total).
- Verify PC0 to PC1 ping across 3 hops + simulation Successful.



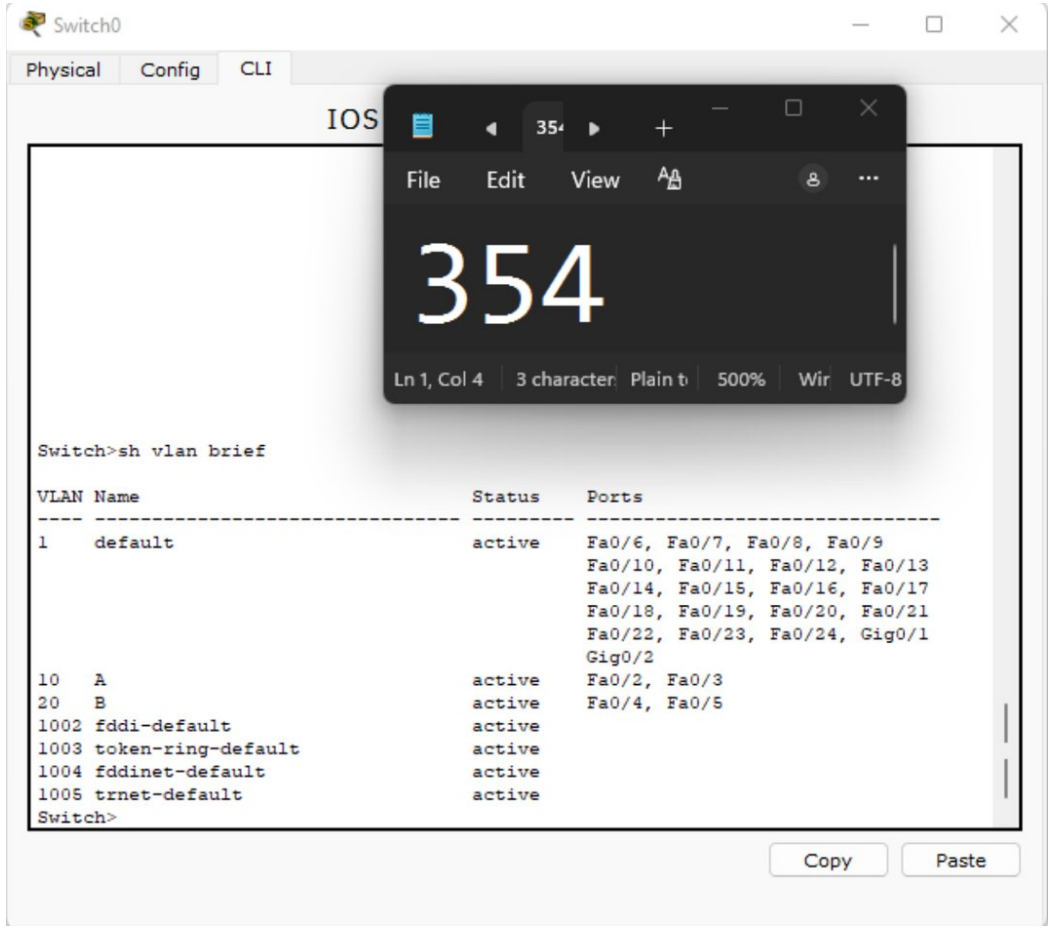
Mixed Ethernet + serial network

- Same static-routing logic, now with serial links (s2/0, s3/0) plus FastEthernet.
- tracert shows each hop.



VLANs with router-on-a-stick

- VLANs split one switch into virtual LANs, e.g. VLAN 10 = A, VLAN 20 = B.
- PC ports = access mode (one VLAN); switch-to-router port = trunk (carries all).
- Router uses subinterfaces (fa0/0.10, .20) with encapsulation dot1q + one IP per VLAN.
- Same-VLAN PCs switch directly; cross-VLAN goes through the router.
- DHCP variant: pools per VLAN + excluded-address for router IPs; PCs set to DHCP.



Checks used in every lab

- ping <ip/name>: IP first, then name (separates connectivity from DNS).
- tracert: lists each router hop; ipconfig: my own address; arp -a: learned MACs.
- Simulation mode: watch ICMP/ARP hop by hop; Successful = delivered.

